

Advancing Security and Trust in WSNs: A Federated Multi-Agent Deep Reinforcement Learning Approach

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Abstract—Wireless Sensor Networks (WSNs) show significant potential through their ability to collect and analyze real-time data, notably enhancing various sectors. The new emerging security threats present a severe risk to the security and reliability of WSNs. Data-driven Artificial Intelligence (AI) leverages WSNs data to deal with new emerging threats like zero-day attacks. However, AI-based models suffer from poor adoption due to the lack of realistic/up-to-date attack data. Recently, Multi-Agent Deep Reinforcement Learning (MARL) has gained significant attention for enhancing Intrusion Detection Systems (IDS) capabilities. MARL offers improved flexibility, efficiency, and robustness. However, this requires data sharing, leading to network bandwidth consumption and slower training. Additionally, the curse of dimensionality hampers its benefits, given the exponential expansion of the state-action space. Privacy-aware collaborative methods such as Federated Learning (FL) emerge as a new approach, enabling decentralized model training across a network of devices while preserving the privacy of each participant. In this context, we introduce a novel framework (MAF-DRL) that leverages FL and MARL to efficiently detect WSN-based attacks. MAF-DRL enables distributed learning across multiple agents with adaptive, flexible, and robust attack detection. We also introduce a trust-based scheduling mechanism that dynamically allocates resources based on agent reliability. This trust-aware approach allows FL systems to adapt to changing network conditions and device behaviors. By prioritizing reliable devices, our method improves the energy efficiency of WSNs and enhances the resilience and effectiveness of the distributed FL paradigm. Finally, we assess the robustness of our framework by testing it against real-world WSN attacks. This evaluation demonstrates its efficiency for secure and communication-efficient federated edge learning across various agents.

Index Terms—Wireless sensor networks (WSNs), E-commerce, multi-agent federated learning, trust evaluation, energy efficiency.

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I. INTRODUCTION

WIRELESS Sensor Networks (WSNs) play a crucial role in enhancing various aspects of e-commerce operations. These networks consist of spatially distributed autonomous sensors/nodes that monitor physical or environmental conditions and transmit data wirelessly [1], [2], [3]. These nodes collaborate to monitor and collect data from their surrounding environments and transmit these data to a central location or a sink node for further processing. WSNs have demonstrated versatility in various applications, including resource management and e-commerce [4]. However, they face challenges related to resource limitations and security [5], [6].

Context-aware WSNs offer a promising solution by tailoring feedback and recommendations to user needs, thereby streamlining daily activities. In this context, the integration of complex object networks holds the potential to enrich user experiences by adapting to dynamic contexts and preferences. Despite these advances, the overarching challenge of resource consumption persists, impacting operational efficiency and scalability. As e-commerce systems evolve, security becomes a paramount concern. Beyond usability and manageability challenges, ensuring robust security measures becomes critical to safeguarding sensitive user data and maintaining trust in online transactions. In this context, several Artificial Intelligence (AI) based solutions have been proposed to detect and predict attacks on these systems [7], [8]. Using the power of AI algorithms, e-commerce platforms can proactively identify anomalous behavior and mitigate potential threats.

Recently, Multi-Agent Deep Reinforcement Learning (MARL) aims at enhancing Intrusion Detection Systems (IDS) capabilities by offering unparalleled flexibility, efficiency, robustness, adaptability, and scalability. Its ability to train multiple agents to collaborate or compete in complex environments enables rapid adaptation to evolving threats. By learning directly from raw data, MARL-based IDS systems automate feature extraction, enhancing detection accuracy while demonstrating resilience against adversarial attacks. The decentralized nature of MARL ensures fault tolerance and redundancy, guaranteeing uninterrupted operation even in dynamic network environments. Furthermore, MARL enables continuous learning, allowing IDS to stay ahead of emerging threats. Its scalability makes it suitable for large-scale network environments, ensuring consistent performance as networks

grow. Despite its advantages, MARL encounters challenges that include data-sharing requirements leading to increased network bandwidth consumption and slower training speeds. Moreover, the curse of dimensionality poses a significant obstacle, as the exponential growth of the state-action space limits the effectiveness of MARL-based approaches.

Privacy-aware collaborative methods such as Federated Learning (FL) emerge as a new approach, enabling decentralized model training across a network of devices while preserving the privacy of each participant. The FL allows for model training across a network of decentralized devices, exemplified by drones operating at the network edge. Within FL, each device autonomously refines its model using locally collected data, focusing on features relevant to its distinct operational context. Significantly, FL ensures data privacy by transmitting only aggregated model updates. This approach not only safeguards sensitive information, but also protects the autonomy of individual devices/sensors [3], [9], [10], [11], [12]. WSNs, characterized by their distributed nature and resource constraints, require FL models that account for the reliability of the device to ensure the integrity and efficiency of the learning process [13]. Thus, it is important to consider the trustworthiness of the sensors and to be context-aware to address the unique challenges posed by WSNs. By incorporating trustworthiness metrics into FL frameworks, we can mitigate the impact of unreliable devices on model accuracy and overall system performance. This proactive approach not only enhances the security and privacy of WSNs but also promotes energy efficiency and resource optimization. Therefore, by prioritizing device reliability and context awareness, we can unlock the full potential of FL in ensuring the integrity and efficiency of machine learning processes within WSNs. The continuous communication demands of FL also contribute to increased network load, and bottlenecks may exacerbate this situation, potentially leading to congestion and straining the network infrastructure. While previous FL studies have explored scheduling techniques aimed at optimizing resource utilization, focusing on resources such as time and local computing power, these approaches often overlooked crucial learning-related factors, including device trustworthiness.

Sensors in WSNs exhibit various behaviors influenced by various factors such as environmental conditions, user interactions, and network activity. In WSNs, sensors can demonstrate different patterns based on their locations, sensing capabilities, and communication protocols. Similarly, in e-commerce systems, sensors embedded in devices or infrastructure can exhibit different behaviors depending on user preferences, purchasing habits, and browsing activities. Understanding and analyzing these behavioral patterns are crucial for detecting anomalies, identifying potential security threats, and optimizing system performance. Therefore, it's crucial to group sensors with similar behaviors into clusters to improve security and enhance overall system efficiency.

To address the challenges outlined above, this paper introduces a novel framework (MAF-DRL) that leverages FL and MARL to efficiently detect WSN-based attacks. Our proposed framework enables distributed learning while ensuring adaptive, flexible, and robust attack detection. Our

proposed framework involves clustering sensors based on their behavior, which allows for more focused monitoring and analysis. This method expedites the detection of suspicious activities or deviations from normal behavior. Additionally, we propose a trust-based scheduling mechanism, to dynamically allocate resources based on agent reliability. This incorporation of trust-awareness enables FL systems to adapt to evolving network conditions and device behaviors. By prioritizing devices with proven reliability, our approach not only enhances the energy efficiency of WSNs but also strengthens the resilience and effectiveness of distributed FL paradigms. Finally, we evaluate the robustness of our framework using real-world WSN attacks. This evaluation demonstrates its efficiency for secure and communication-efficient federated edge learning across various agents.

The main contributions of this paper are summarized as follows:

- We propose a novel framework (MAF-DRL) that leverages FL and MARL to efficiently detect WSN-based attacks. MAF-DRL leverages MARL and FL for intrusion detection within WSNs, empowering autonomous identification and response to potential security threats.
- We design a novel strategy of clustering sensors based on behavior, enabling focused monitoring and analysis for expedited detection of suspicious activities.
- We design a scheduling algorithm that meticulously allocates necessary bandwidth to reliable sensors, emphasizing the reduction of communication expenses during the learning process and prioritizing participants demonstrating superior model learning performance.
- We evaluate the effectiveness of our framework by testing it against real-world WSN attacks.

The rest of this paper is organized as follows: Section II reviews the current state of the art of WSN-based attacks. Section III outlines our system model. Section IV introduces our MAF-DRL framework. Section V details the trust-based recommendation system. Section VI presents the performance evaluation. Finally, Section VII concludes the paper.

II. RELATED WORK

WSNs have demonstrated their adaptability in numerous fields [14], [15]. WSN security has become crucial, with several state-of-the-art schemes leveraging AI techniques to mitigate attacks in the wireless networks. In this section, we delve into some of the most prominent schemes leveraging AI techniques to ensure WSN security, along with their limitations.

Vu et al. [16] focused on the design and implementation of a Mobile Ad Hoc Network (MANET) routing protocol optimized for energy efficiency within 5G networks. Khera et al. [17] evaluated existing clustering-based methodologies aimed at improving energy efficiency in sensor networks. The authors introduced a novel method called Hibernated Clustering Wireless Sensor Networks to address these shortcomings. This methodology focused on extending the operational duration of individual sensor nodes, thereby enhancing the overall network lifespan. Liu et al. [18]

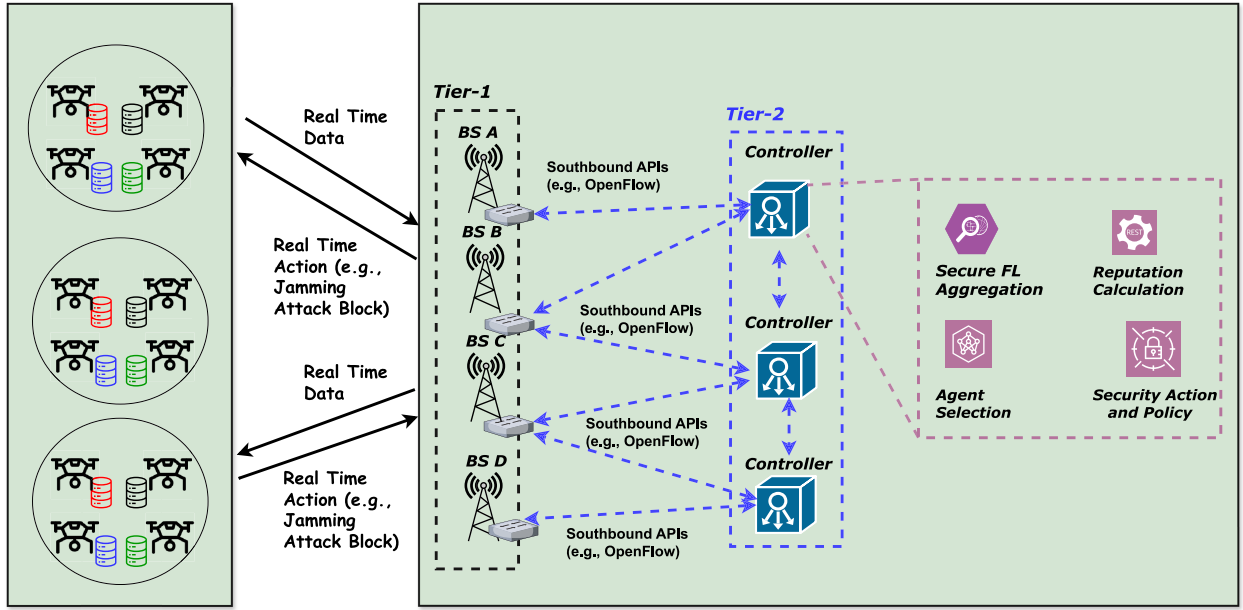


Fig. 1. Two-Tier System Architecture for Enhanced WSN Security and Efficiency.

proposed a routing protocol modification, termed the Improved Energy-Efficient LEACH (IEE-LEACH) protocol. The IEE-LEACH protocol takes into consideration both the remaining energy levels of individual nodes and the average energy status of the entire network. Wei et al. [19] introduced a distributed clustering algorithm to adapt cluster sizes based on the hop distance to the data sink.

The importance of trust spans across WSNs and e-commerce platforms, where it ensures reliability and integrity. In WSNs, trust guarantees accurate data exchange, while in e-commerce, it fosters secure transactions. Chen et al. [20] introduced IoTrust, a novel trust framework that integrated Software Defined Network (SDN) with the Internet of Things (IoT), alongside a cross-layer authorization protocol rooted in IoTrust principles. This innovative approach provided a unique angle on trust administration within IoT systems. Kumar et al. [21] introduced a novel approach to evaluate trust in WSNs by employing ML techniques. This method involves identifying specific trust attributes using a predefined model and then devising a unique algorithm that utilizes machine learning to categorize and integrate these attributes into a comprehensive trust value. Chen et al. in [20] introduced a novel approach to assess user trust within social e-commerce platforms, aimed at addressing the shortcomings of conventional methods marked by inaccurate evaluations and inefficiencies. This method leveraged AI techniques to establish a novel framework grounded on diverse factors such as business, environmental, and social elements, facilitating user trust levels.

Security measures are indispensable for safeguarding data integrity and confidentiality in WSNs. Kumar et al. [21] introduced an innovative approach termed Intelligent Clustering for Intelligent Transportation Systems (ICITS), which employed a unique selection method for Cluster Heads (CHs) called GABAT, blending Genetic Algorithm (GA) and BAT

Algorithm. ICITS aimed to meet stringent security and reliability requirements while collecting data from sensor nodes deployed in road transport scenarios. Yingzi and Zhaoji [22] proposed a hierarchical trust management system, TSW, tailored for Secure and Dynamic Wireless Sensor Networks (SDWSNs).

To address the issues of these existing solutions, we introduce a novel framework (MAF-DRL) that leverages FL and MARL to efficiently detect WSN-based attacks. Our proposed framework enables distributed learning while ensuring adaptive, flexible, and robust attack detection. Our proposed framework involves clustering sensors based on their behavior, which allows for more focused monitoring and analysis. This method expedites the detection of suspicious activities or deviations from normal behavior. Additionally, we propose a trust-based scheduling mechanism, to allocate resources based on agent reliability dynamically. This incorporation of trust-awareness enables FL systems to adapt to evolving network conditions and device behaviors. By prioritizing devices with proven reliability, our approach not only enhances the energy efficiency of WSNs but also strengthens the resilience and effectiveness of distributed ML paradigms.

III. SYSTEM ARCHITECTURE

Our system architecture (see Fig. 1) is structured into two tiers: Tier-1 consists of Base Stations (BS A, BS B, BS C, BS D) responsible for collecting and processing real-time data at the edge layer. Tier 2 comprises controllers to manage and process data from Tier 1. These controllers handle various critical functions, including the aggregation of secure FL data, reputation calculation, agent selection, and implementation of security actions and policies. The controllers communicate using Southbound APIs, such as OpenFlow, to ensure real-time data exchange and immediate response actions, such as

blocking jamming attacks. This architecture facilitates distributed learning across multiple agents with adaptive, flexible, and robust attack detection, leveraging both FL and MARL to improve the security and efficiency of WSNs. Table I shows the notations used to present our system model. Also, our proposed system model allows:

- 1) **Agent Representation:** Each node in the network acts as an agent, representing either an individual device or a cluster of devices. Agents maintain localized datasets containing pertinent information about network activities, system logs, and relevant features.
- 2) **Local Attack Detection Models:** Agents independently maintain their local attack detection models. Deep Q-Learning (DQL) is employed as the primary method for attack detection in this decentralized setup. DQL enables agents to continuously refine their models by dynamically interacting with the environment, resulting in a robust and adaptive defense mechanism against potential threats. The iterative improvement of local attack detection capabilities enhances the collective intelligence of the system, fostering a comprehensive and resilient security posture.
- 3) **Collaborative Learning:** Agents participate in collaborative learning via our FL training process, collectively contributing to the refinement of the global attack detection model. Through this collaborative effort, agents leverage their diverse datasets to enhance the effectiveness and robustness of the global attack detection model.
- 4) **Recommendation System based on Reputation:** Each agent has a reputation score that reflects the trustworthiness and reliability of individual devices within the network. By considering reputation scores, the recommendation system can offer personalized and reliable recommendations tailored to the specific needs and preferences of users. This enhances the overall user experience and fosters a collaborative and trustworthy environment within the WSN ecosystem.
- 5) **Resource Usage Personalization:** We introduce an innovative scheduling algorithm that prioritizes participants with higher reputation scores, ensuring efficient utilization of resources. By strategically allocating bandwidth to trusted devices, we aim to enhance the overall efficiency of the federated learning system, reducing overhead while accelerating model convergence.

IV. SECURE AND EFFICIENT MULTI-AGENT FEDERATED LEARNING

The traditional update rule in Q-learning for a single-agent DQL is given by:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left(r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right) \quad (1)$$

We then adapt the update rule for multiple agents, where i represents the agent's index:

$$Q_i(s, a_i) \leftarrow Q_i(s, a_i) + \alpha \left(r_i + \gamma \max_{a'_i} Q_i(s', a'_i) - Q_i(s, a_i) \right) \quad (2)$$

TABLE I
NOTATIONS

Notations	Definition
$V_{\text{FJ}}(s)$	Federated joint value function for the state s .
$Q_{\text{FJ}}(s, a)$	Federated joint action-value function for the state s and joint actions a .
$R_{\text{detection}}$	The reward for attack detection.
α	Learning rate.
$\nabla_{x_{k,j}}$	The gradient with respect to $x_{k,j}$.
N	The number of collaborators/participants.
Y_i	The number of local data samples of participant i .
$r_{\max}$	The maximum number of training rounds.
$\text{clip}(\cdot)$	Ta function that restricts the values to fall within the range $[0, 1]$.

Then, we introduce the federated joint value function, which quantifies the expected cumulative return when a group of agents collaborates to detect and mitigate attacks within a shared network:

$$V_{\text{FJ}}(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \sum_{i=1}^N r_i^t \mid s_0 = s \right] \quad (3)$$

where, $V_{\text{FJ}}(s)$ represents the federated joint value function for the state s , while N is the number of agents dedicated to attack detection and mitigation. In this setting, $s_0 = s$ denotes the initial state, t denotes the time step, and r_i^t denotes the reward received by agent i at time step t .

Furthermore, we define the federated joint action-value function for attack detection and mitigation as the cumulative return when all agents collaboratively take a set of joint actions to detect and mitigate attacks in a shared environment using federated learning.

$$Q_{\text{FJ}}(s, \mathbf{a}) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \sum_{i=1}^N r_i^t \mid s_0 = s, \mathbf{a}_0 = \mathbf{a} \right] \quad (4)$$

where, $Q_{\text{FJ}}(s, \mathbf{a})$ signifies the federated joint action-value function for the state s and joint actions $\mathbf{a}$, with N as the number of agents engaged in the detection and mitigation of attacks using FL (see Algorithm 1).

A. Cluster-Based Anomaly Detection and Reward Optimization

In the pursuit of enhancing anomaly detection within the Multi-Agent Framework for Deep Reinforcement Learning (MAF-DRL), we introduce an innovative approach that leverages clustering techniques to group agents exhibiting similar behaviors. This method significantly improves anomaly detection by focusing on deviations or unusual patterns within these clusters. By isolating and analyzing specific cluster behaviors, the detection and response to security attacks become more effective. We formulated this problem as an optimization

Algorithm 1 Multi-Agent Federated Deep Reinforcement Learning (MAF-DRL)**Input:** Network environment, sensor agents, federated learning parameters**Output:** Collaborative reinforcement learning model

```

1: Initialize federated learning environment
2: Initialize sensor agents with DQL models  $\theta_i$ 
3: Initialize federated learning parameters: learning rate  $\alpha$ ,
   discount factor  $\gamma$ , exploration rate  $\epsilon$ 
4: for each iteration  $t$  do
5:   for each sensor agent  $i$  do
6:     Receive state observation  $s_i$  from environment
7:     Select action  $a_i$  using DQL model:  $a_i = \arg\max_a Q(s_i, a; \theta_i)$ 
8:     Execute action  $a_i$  in environment
9:     Receive reward  $r_i$  from environment
10:    Update DQL model using Eq. (3)
11:   end for
12:   Share updated DQL models with other agents
13:   Aggregate models to form global model:  $\bar{\theta} \leftarrow \frac{1}{N} \sum_{i=1}^N \theta_i$ 
14: end for
15: return Collaborative reinforcement learning model  $\bar{\theta}$ 

```

problem in which the objective is to maximize the cumulative reward attained by the selected agents during training. The reward for attack detection ($R_{\text{detection}}(S_t)$) incentivizes agents to accurately identify security threats based on the current state. It is determined by a weight factor $w_{\text{detection}}$ and can be expressed as:

$$R_{\text{detection}}(S_t) = \begin{cases} 1 & \text{if an attack is detected} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

Each agent's participation in the training process is represented by a binary variable ($x_{k,j}$), where 0 denotes non-selection and 1 denotes selection. This problem is formulated as:

$$\underset{x}{\text{maximize}} \quad \sum_{t=0}^{\infty} R(s_t, \mathbf{a}_t) x_{k,j} \quad (6a)$$

subject to

$$\sum_j x_{k,j} \leq 1, \quad \forall k \in [1, K], \quad (6b)$$

$$\sum_k x_{k,j} \geq 1, \quad \forall j \in [1, J], \quad (6c)$$

$$x_k \in \{0, 1\}, \quad \forall k \in [1, K]. \quad (6d)$$

where, $x_{k,j}$ represents the binary variable indicating agent selection for training, subject to constraints ensuring each cluster has at least one agent (6b) and each agent belongs to only one cluster (6c). The binary nature of the decision variable is enforced by constraint (6d). Notably, Problem (6a) is NP-hard due to its mixed-integer non-linear nature.

To tackle this complexity, we propose employing stochastic gradient descent (SGD), where devices are assigned to clusters satisfying the constraints. The problem is then converted

into a differentiable form suitable for SGD, allowing for gradient-based optimization. This transformation enables efficient solution space exploration, facilitating the maximization of total reward values for the selected agents during training.

Subsequently, we employ SGD to iteratively update the values of $x_{k,j}$ to maximize the objective function. The update rule for each iteration i can be defined as follows:

$$x_{k,j}^{(i+1)} = \text{clip} \left(x_{k,j}^{(i)} + \alpha \nabla_{x_{k,j}} \sum_{t=0}^{\infty} R(s_t, \mathbf{a}_t) x_{k,j}^{(i)}, 0, 1 \right) \quad (7)$$

where, α denotes the learning rate, $\nabla_{x_{k,j}}$ represents the gradient concerning $x_{k,j}$, and $\text{clip}(\cdot)$ is a function that restricts the values to fall within the range $[0, 1]$.

V. RECOMMENDATION SYSTEM BASED ON TRUST

In our proposed MAF-DRL, we utilize Graph Neural Networks (GNNs) to enhance the evaluation of participants' trust in the FL process. Our proposed approach leverages a graph structure to model the relationships and interactions among participants. Also, we introduce a consensus-based trust algorithm to capture the variance or disagreement among the FL participants. In this context, we use a distributed linear iterative algorithm to achieve consensus in the network.

Let $G = (P, E)$ be a network of FL participants, where P represents the set of FL participants and E represents the set of directed edges (connections between participants).

Each FL participant $p \in P$ has an initial state $x(0)_p$, which is a real number representing its current value or opinion. The objective is to achieve consensus among the participants, where all participants eventually converge to a common value. The distributed linear iterative algorithm can be described by the following dynamics:

$$x(t+1)_p = \sum_q w(p, q) x(t)_q \quad (8)$$

where, $x(t+1)_p$ denotes the state or feedback of participant p at time $t+1$, and $x(t)_q$ denotes the state or feedback of participant q at time t . The term $w(p, q)$ represents the weight of the participant directly connected to p .

The algorithm then updates the state of each participant at each time step by taking a weighted average of the states of its neighboring agents. The weights $w(p, q)$ determine the influence or importance of each neighbor in the update process. In cases where participants p and q do not communicate with each other, the weights are $w(p, q) = 0$. In such instances, the FL participants converge *i.e.*, $\lim_{t \rightarrow \infty} x(p) = x(\infty)$.

Based on the feedback of other FL participants, we evaluated each participant's trust score. We averaging the feedback ratings $T(p)$ for participant p as follows:

$$T(p) = \frac{\sum_{q=0}^N x(t+1)_q}{N} \quad (9)$$

where, $x(t+1)_q$ represents the FL participants providing feedback about participant p , and N is the total number of participants providing feedback.

We then assign weights as follows:

$$T(p) = \frac{\sum_q w(q, p) x(t+1)_q}{\sum_q w(q, p)} \quad (10)$$

where, q represents the FL participants providing feedback about p .

A. Behavioral Clustering for Enhanced Anomaly Detection

The level of agreement or convergence achieved by participants within the network is assessed using a consensus error metric, which quantifies the disparity between the actual states of participants and the desired consensus value. Typically, the consensus value is determined as the average state of all participants within the network. The difference between a participant's actual state and the consensus value is termed as the consensus error, reflecting the reliability of FL participants.

We define the reliability vector $x(t) = [x_1(t), x_2(t), \dots, x_n(t)]$ at time t , where $x(t)_p$ denotes the reliability state of participant p at time t . Here, the reliability state is binary, with $x(t)_p = 1$ indicating a reliable agent and $x(t)_p = 0$ indicating an unreliable agent.

The consensus error at time t quantifies the discrepancy or deviation of each participant's state from the average state. To represent the consensus error at time t , we use the root mean square error, defined as:

$$\text{ConsensusErr}(t) = \sqrt{\frac{1}{N} \sum_{p=1}^N (x(t)_p - \bar{x}(t)_p)^2} \quad (11)$$

where, $\bar{x}(t)_p$ is the average state of all participants at time t .

The trust score is calculated based on the consensus error, where participants with smaller error values are considered more trustworthy. To provide a thorough assessment of a participant's trust, we accumulate the trust scores over multiple time steps. Formally, the trust of a participant p can be expressed as:

$$T'(p) = \frac{1}{T} \sum T(p, t) \quad (12)$$

where $T(p, t)$ represents the trust of participant p from time $t = 1$ to $t = T$.

Incorporating feedback alongside the consensus error allows for a more nuanced evaluation. While the consensus error measures the level of agreement within the network, the feedback score captures the direct feedback received by participants. The global trust of a participant is then computed as a combination of these factors:

$$T''_p = \alpha x(p) + \beta \left(\frac{1}{\text{ConsensusErr}(p)} \right) \quad (13)$$

where $x(p)$ denotes the feedback-based trust received for participant p , $\text{ConsensusErr}(p)$ is the consensus error of participant p , and α and β are weighting variables that regulate the influence of feedback and consensus in trust calculation. These components can be weighted based on their perceived relevance or trustworthiness, similar to the feedback-based trust formulation.

We then define the global trust of an FL participant as:

$$T''(p) = (W_{\text{feedback}} \times x(p)) + \left(W_{\text{consensus}} \times \frac{1}{\text{ConsensusErr}(p)} \right) \quad (14)$$

Algorithm 2 Consensus Trust Algorithm

```

1: Input: Feedback from participants
2: Output: Consensus trust score for the target participant
3: Initialization:
4: Initialize trust scores for all participants to neutral values
5: Specify the number of iterations and consensus threshold
6: Feedback Collection:
7: Gather feedback about the target participant's behavior
   from multiple sources
8: Consensus Error Trust Calculation:
9: for  $i = 1$  to  $N$  do
10:   Compute the consensus error trust using Equation (8)
11:   Update the trust scores of all participants using
   Equation (11)
12: end for
13: Consensus Verification:
14: if  $T''(p) \leq \frac{\sum_{i=0}^N T''(i)}{N}$  then
15:   return Consensus Reached
16: else
17:   return Consensus Unreached

```

where W_{feedback} and $W_{\text{consensus}}$ represent the weights assigned to feedback-based trust and consensus-based trust, respectively.

To verify the trustworthiness of a participant, we propose a consensus algorithm among all participants. Algorithm 2 calculates the consensus trust score for the target participant using inputs from various participants. Initially, it sets the number of iterations and the consensus threshold, then initializes a neutral trust value for each participant. The algorithm collects information on the behavior of the target participant from multiple sources and computes the difference between each participant's trust score and the overall average trust score. It updates all participants' trust scores using the consensus error values. Finally, consensus is determined by checking if the average trust score of each participant differs from the target participant's trust score by less than the consensus threshold.

B. Agent Selection

Agent selection is crucial in dynamic environments where diverse skills are required for each agent to perform effectively. Optimal agent selection improves overall system performance and reduces computational complexities, thus improving the efficiency of the learning process. Given the resource-intensive nature of large-scale models, efficient scheduling is necessary to evenly distribute the workload and optimize learning efficiency. We propose an agent scheduling approach focused on resource optimization to prevent delays or interruptions during model transmission. This study utilizes Orthogonal Frequency Division Multiple Access (OFDMA), which allows simultaneous data transmission from multiple agents by assigning distinct subcarrier subsets to individual users, ensuring concurrent and interference-free communication.

The transmission rate (r_k) of a device to the Mobile Edge Computing (MEC) server is calculated as follows:

$$r_k = \alpha_k B \log_2 \left(1 + \frac{P_k g_k}{\alpha_k B N_0} \right), \quad \forall k \in [1, K], \quad (15)$$

Algorithm 3 Greedy Bandwidth Allocation Algorithm

```

1: Input: Total_Bandwidth,  $T$ ,  $t_{\text{train}}$ , and  $t_{\text{up}}$ 
2: Output:  $x_k$  and  $b_k$ 
3:  $N \leftarrow \text{length}(T)$ 
4: remaining_bandwidth  $\leftarrow$  Total_Bandwidth
5: selected_agents  $\leftarrow$  empty_list()
6: sorted_agents  $\leftarrow$  SORTAGENTSBYTRUST( $T$ )
7: for each agent in sorted_agents do
8:    $k \leftarrow$  agent.index
9:   if isCandidate( $k$ )  $\wedge$  has Sufficient Bandwidth( $k$ ) then
10:     select Agent( $k$ )
11:     allocate Bandwidth( $k$ )
12:     update Remaining Bandwidth( $k$ )
13:     add To Selected Agents( $k$ )
14:   end if
15: end for
16: return  $x_k, b_k = 0$ 

```

where r_k represents the transmission rate of device k to the MEC server, α_k denotes the allocated bandwidth for device k , B signifies the total available bandwidth, P_k denotes the transmission power of device k , g_k represents the channel gain of device k , and N_0 stands for the power spectral density of Gaussian noise.

We then propose a device scheduling algorithm that optimizes not only the overall trust scores but also additional factors related to agents' interactions and the learning process.

The training time (t_k^{train}) and upload time (t_k^{up}) are calculated as follows:

We estimate the training time as:

$$t_k^{\text{train}} = \epsilon |D_k^t| \frac{\zeta_k}{f_k} \quad (16)$$

where ϵ serves as a scaling factor, D_k^t represents the dataset of agent k , ζ_k denotes the number of CPU cycles required to process a single data sample, and f_k represents the processing power of the CPU.

The upload time for transmitting a model from agent k to the central server can be expressed as:

$$t_k^{\text{up}} = \frac{s}{r_k} \quad (17)$$

where s denotes the size of the model to be uploaded, and r_k signifies the achievable data rate of agent k .

We formulate an optimization problem for device scheduling that incorporates bandwidth constraints into the decision-making process. The objective is to maximize the cumulative reward, denoted by $\sum_k R_k x_k$, where x_k is a binary variable indicating the selection of agent k . The optimization problem is expressed as:

$$\underset{x, b}{\text{maximize}} \quad \sum_k R_k x_k \quad (18a)$$

subject to

$$\sum_k b_k \leq 1 \quad (18b)$$

$$x_k (t_k^{\text{train}} + t_k^{\text{up}}) \leq T, \forall k \in [1, K] \quad (18c)$$

$$x_k \in \{0, 1\}, \forall k \in [1, K] \quad (18d)$$

$$0 \leq b_k \leq 1, \forall k \in [1, K] \quad (18e)$$

where, b_k denotes the bandwidth fraction allocated to agent k . The constraint (18b) ensures that the total bandwidth allocations among all agents do not exceed the available bandwidth. In our formulation, where timely responses are critical and operations must align with specific temporal requirements dictated by the dynamic WSN environment, we assert in constraint (18c) that the selected agents must complete their model training and upload within a predefined time frame. Constraint (18d) denotes the binary nature of x_k , designating device k as either selected or not. Bandwidth allocation ratios are bounded by constraint (18e). In particular, considering constraints (18b), (18c), (18d), and (18e) solely mirrors the complexity of a knapsack problem, making it NP-hard. To tackle this challenge, we employ a greedy method, as detailed in Algorithm 3, effectively navigating the constraints while optimizing agent selection and bandwidth allocation, recognizing the intricate interplay between bandwidth considerations and model training deadlines.

VI. PERFORMANCE EVALUATION

We implemented our proposed framework (MAF-DRL) using PySyft, a privacy-aware deep learning framework built on the PyTorch platform. We have considered a test environment, where multiple agents were deployed to conduct local training and then through the FL process conduct collaborative learning in a fully distributed manner. This capability empowers every agent to effectively attain an optimal policy, maximizing the cumulative expected reward, while simultaneously upholding the privacy of each agent's data. Our exploration encompasses two distinct scenarios (see Table II and Table III). The initial scenario involves binary classification, where the focus is on binary classification for jamming events. The trained model is specialized in distinguishing between two possible outcomes: the presence or absence of jamming interference. This approach proves valuable in scenarios where a simple binary decision is sufficient, such as determining whether a communication channel is under intentional disruption. In the second scenario, we extend our approach to multiclass classification for jamming attack detection. This involves configuring the model to handle more than two classes, allowing it to categorize different types of jamming attacks. This classification is particularly useful in sophisticated environments where various jamming techniques may be employed, each requiring a specific response. Throughout both scenarios, the training process involves optimizing the model's parameters to discern patterns and features indicative of jamming activities. The objective is to enhance the model's accuracy in detecting and classifying jamming events, contributing to the overall resilience of the Wireless Sensor Network.

To feed the initial environment with attack data, we used the WSN-DS dataset, which includes four distinct jamming attacks that were simulated using the LEACH protocol. These attacks comprise black hole, gray hole, flood, and scheduling attacks. In the black hole attack, a malicious node masquerades

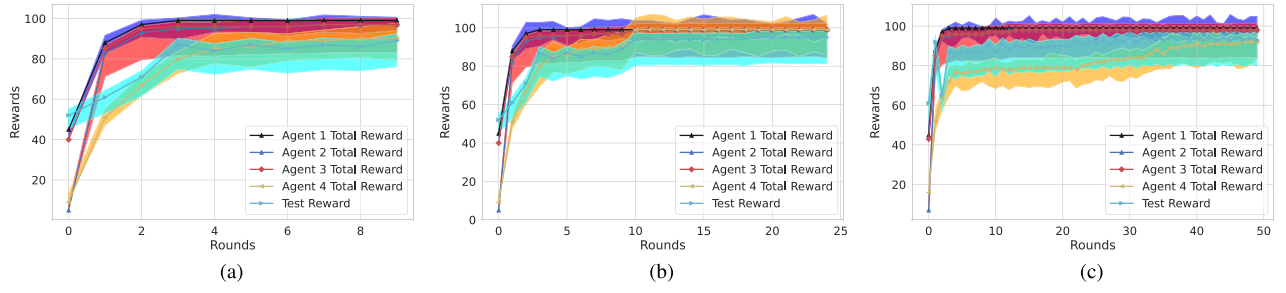


Fig. 2. Agent rewards for (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.

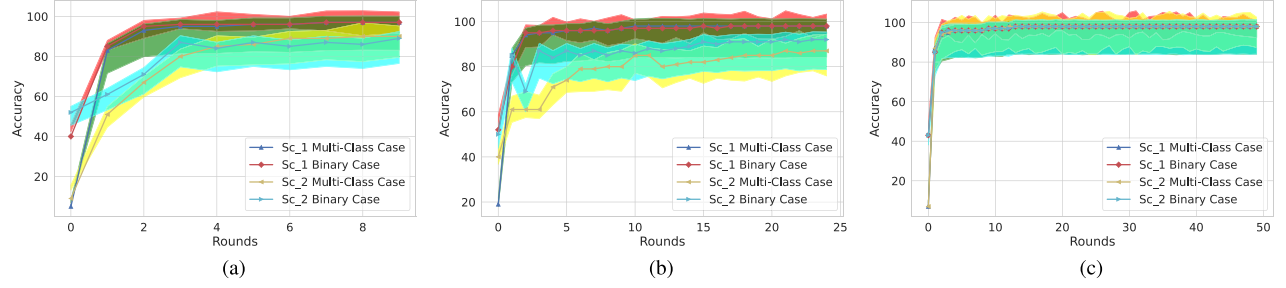


Fig. 3. Model Accuracies for (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.

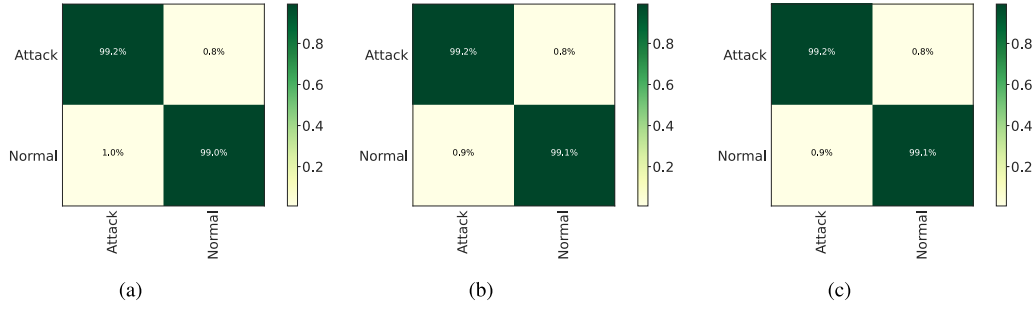


Fig. 4. Confusion matrices for (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.

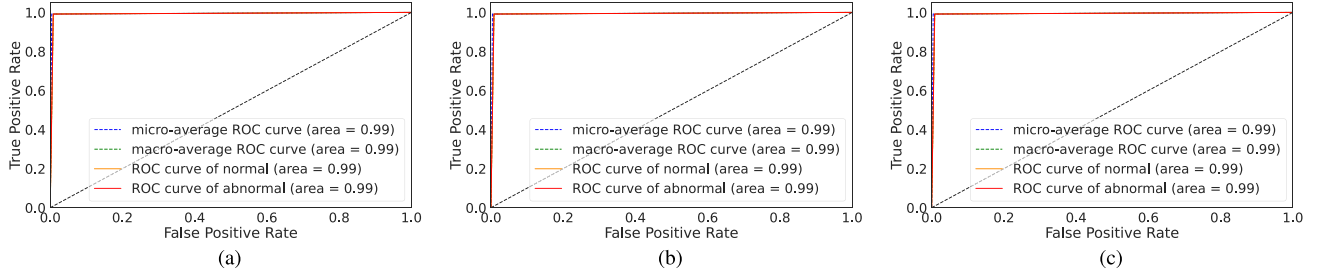


Fig. 5. ROC Curves for (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.

as a Cluster Head (CH) and intentionally discards all data packets relayed through it. The Grayhole attack involves a deceptive CH selectively dropping packets, introducing a random or targeted element. Flooding attacks disrupt the LEACH protocol by flooding it with numerous advertising CH messages, leading to increased energy consumption and delayed decision-making by nodes. Scheduling attacks take place during the setup phase, causing data loss by assigning the same time slot to all cluster members to transmit data. The attacks were simulated with varying intensities (10%, 30%, and 50%) to thoroughly assess their impact on the

lifetime of the WSN, energy consumption, and overall network services.

In this simulated environment, various scenarios were explored from 10 to 50 rounds of training. The evaluation of our proposed framework involved an extensive examination using key metrics, including accuracy, F1 score, receiver operating characteristic (ROC) curves, and confusion matrices. These metrics serve as crucial indicators for quantifying the model's performance across different dimensions.

Figs. 2(a), 2(b), and 2(c) show the convergence (*i.e.*, cumulative average rewards) of our framework over 10, 25, and

TABLE II
WSN-DS LABELS

Label	Multi-class	Binary
Normal	0	0
Blackhole	1	1
Flooding	2	1
Grayhole	3	1
Scheduling (TDMA)	4	1

TABLE III
CLASSIFICATION SCENARIOS

Scenarios	Type	Rounds	Local epochs
1	Binary Classification Case	10,25,50	1-5
2	Multi-class Classification Case	10,25,50	1-5

50 training rounds, respectively. We observe a continuous improvement in the average reward with increasing training rounds, reaching its maximum after only three rounds of training. Figs. 3(a), 3(b), and 3(c) show the obtained accuracy in both scenarios for 10, 25, and 50 training rounds, respectively. We observe that the accuracy of our proposed framework increases, reaching a maximum of 99. This shows that the agents learn from each other while preserving privacy, leading to a high accuracy score in the global model. Figs. 4(a), 4(b), and 4(c) show the confusion matrix curves of our framework, highlighting its performance over 10, 25, and 50 training rounds, respectively. Figs. 5(a), 5(b), and 5(c) show the confusion matrix curves of our framework, highlighting its performance over 10, 25, and 50 training rounds, respectively. Our framework achieves an accuracy of 99% and an F1 score of 99%. These curves offer a detailed visual breakdown of the model's classification outcomes, emphasizing the distribution of true positive, true negative, false positive, and false negative instances. These obtained results emphasize the robustness and reliability of our framework, showcasing its ability to produce consistently accurate and precise predictions across varying training rounds. The visual representation through confusion matrix curves provides a general understanding of the model's performance, further reinforcing its effectiveness in the context of the implemented jamming attack detection system.

Table. IV shows the performance metrics of our proposed framework and current AI-based methods using the WSN-DS dataset. We have compared the performance of our proposed framework with some recent AI-based solutions (*i.e.*, Naive Bayes (NB) [23], Multilayer Perceptron (MLP) [24], Adaptive boosting [25], and Boosting [25]) using the same WSN-DS dataset. The remarkable performance metrics, including accuracy, precision, recall, and F1 score all reaching 99%, signify the robustness and reliability of our proposed solution. One of the key highlights of our framework is its ability to achieve such exceptional results with a minimal training time of only 52.12 seconds. This aspect underscores the efficiency and time-saving benefits that our solution brings to the table. The rapid convergence of our model during training further emphasizes its suitability for real-time applications and

TABLE IV
PERFORMANCE METRICS OF MAF-DRL AND CURRENT AI-BASED METHODS

AI algorithm	Accuracy	Precision	Recall	F1	Time (sec-ond)
Naive Bayes	0.72	NA	NA	NA	NA
Multilayer Perceptron	0.7	NA	NA	NA	NA
Adaptive boosting (Adaboost)	0.94	0.84	0.85	0.84	109
Boosting	0.98	0.97	0.97	0.97	2880
Our MAF-DRL	0.99	0.99	0.99	0.99	52.12

resource-constrained environments, such as wireless sensor networks. These promising results not only showcase the prowess of our proposed framework but also position it as a leading solution for WSN-DS applications. The high accuracy, precision, recall, and F1 score collectively validate the framework's ability to accurately classify and detect attacks in the WSN-DS dataset.

VII. CONCLUSION

In this paper, we have introduced a novel framework that uses a Multi-Agent federated learning and deep reinforcement learning Approach to secure WSNs against new emerging threats while preserving privacy. Our proposed framework facilitates collaborative learning among multiple agents. This enables each agent to efficiently acquire an optimal policy that maximizes the cumulative expected reward while maintaining the privacy of individual agent data. The obtained results showed that our proposed framework improves WSN security while ensuring privacy and resilient defense against potential threats in the WSNs ecosystem. For future directions, we intend to enhance the framework's robustness against adversarial attacks by incorporating adversarial training approaches and developing mechanisms to detect and mitigate adversarial manipulation attempts targeting the federated learning process.

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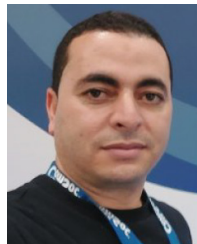
the security of next-generation networks.

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